

NARRATIVE QUARTERLY REPORT

(JULY–SEPTEMBER) 2026

Project — In situ monitoring of water quality in Cuban bays: creating and promoting the use of scientific tools

Participating institutions

- University of Havana (UH)
- Université Libre de Bruxelles (Belgium)

Cooperating institutions

- Center for Environmental Research and Transport Management (CIMAB)
- ESPOLETA Tecnologías (SME)

Other collaborating institutions

- José Antonio Echeverría Technological University of Havana (Cujae)
- Havana Bay State Management Group
- Université Catholique de Louvain

Project coordinators

Cuban side: Dr. C. Ernesto Altshuler Álvarez

Foreign side: Dr. Denis Terwagne

Participants — Dr. Geydis Fundora Nevot, Dr. Florence Degavre, Dr. Stamatios Nicolis, Dr. Alexandre Campo, Lic. Daniela Martínez Ortiz (PhD candidate), Eng. Alex Rivera Rivera (PhD candidate), MSc. Luis Abel Rodríguez de Torner (PhD candidate), Dr. Aramis Rivera Denis, Eng. Orestes Chávez Linares, Dr. Dayaris Hernández.

Duration — Five years

General objective — To provide the Cuban scientific-social ecosystem with a set of tools that enables in situ monitoring of water-quality parameters in Cuban bays, in a sustainable way and connected to local communities.

Specific objectives

- Design and build a wave channel at the Faculty of Physics, UH
- Design and build a portable toolkit for in situ measurement of the water quality of Havana Bay
- Set up a server with a website to receive data
- Establish a methodology for using these tools
- Train the corresponding human resources

Fulfillment of the quarterly objectives

Task 1. Detecting and taking measures against vulnerabilities in the Physics building that endanger the assets associated with the project.

With the arrival of the contents of the first ULB container (detailed below), three cameras were installed covering the three main interior areas where most of the assets acquired with the ARES project are kept. Information cannot be controlled or extracted from them from outside the building. They were installed personally by the project director, with the collaboration of Raciél Pupo-Santos.

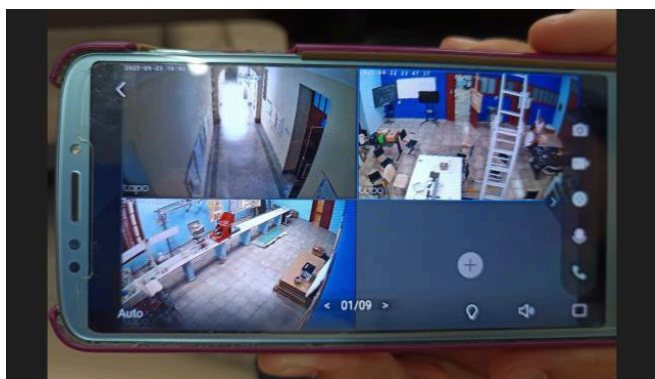


Fig. 1 Trying to improve the security of the premises.

In response to a call from the university leadership for units to report their material needs, the section of the Center for Complex Systems that is directly linked to the ARES project submitted a single request: the urgent need to repair the windows of the rooms where the assets acquired through the project are kept. We are awaiting the corresponding reply.

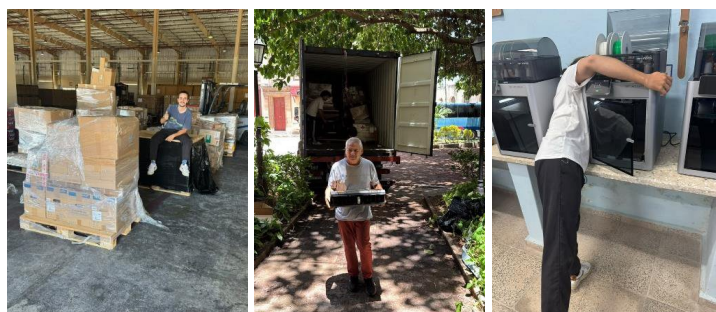
Task 2. Management of goods acquired (and to be acquired) with the project's funding.

As reported in the previous quarterly report, the first container from Brussels arrived at the port of Mariel (Cuba) in April 2026. Given the fuel shortage in the country, the importing company (EMIDICT) did not come to collect the container, even after we met with its officials on June 30 to try to speed up the process. As a result, transportation was financed by the SME “Espoleta Tecnologías”, supplemented by a personal contribution. The equipment arrived at our building on July 15, 2026.

After the contents of the first container arrived, a great deal of work was carried out during the vacation period to install the printers, machine tools, etc., which involved everything from new electrical installations to a detailed inventory-control process.

We continue to monitor the processes for the second container from Belgium and the second shipment from Panama, which are subject to the same high level of internal bureaucracy, despite the worsening situation in terms of shipping companies willing to carry cargo to the country.

The company ALIANC Group Servicios Integrales (<https://www.aliancgroup.com>) and, in particular, the specialist Leonardo Durán, carried out the procedures to finally obtain and install, in September 2026, the spectrophotometer software that we were owed.



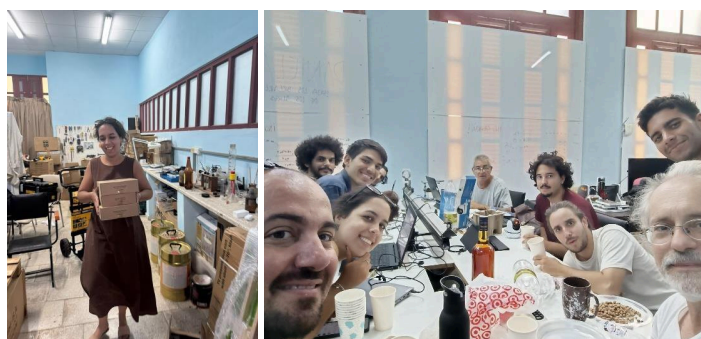


Fig. 2 Arrival of the cargo of the first container from the ULB. Top row, left to right: Alex M. Rivera standing on the unpacked cargo at the port of Mariel; Orestes Chávez carries the equipment to the Physics building from a truck rented by our own means; Alex M. Rivera installing the 3D printers. Bottom row, left to right: Daniela Mtnez. organizing the boxes inside the laboratory; our group celebrating the successful arrival of the container (July 23, 2026).

Task 3. Continuing the instrumentation for in situ measurements.

Given the relatively scarce number of measurements on Havana Bay water samples during the period, owing to various factors, A. Rivera and E. Altshuler took samples at several points of the bay and ran pH, conductivity and UV-visible spectrophotometry analyses on them, using the commercial spectrophotometer acquired with ARES funding (Fig. 3).

They concluded that most of the useful information lies in the UV region of the spectrum, and that measuring turbidity independently could greatly help in analyzing the spectrophotometry data. They also observed logically correlated variations among spectrophotometry, conductivity and pH measurements across different locations in the bay, suggesting pollution levels that can be distinguished from one another using these techniques.

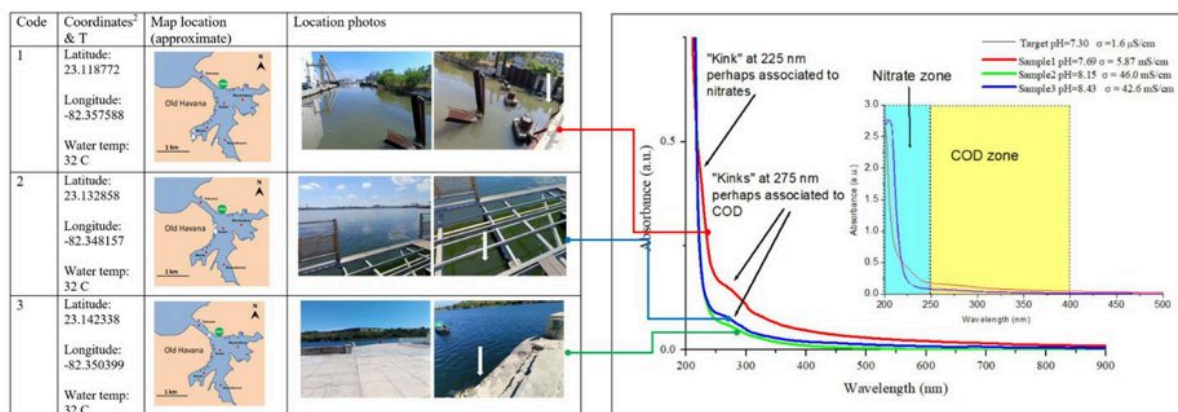


Fig. 3 Study of real water samples from Havana Bay.

Task 5. Start of the construction of the medium-sized wave channel

During this period, the construction of the wave channel received an extraordinary boost. First, fifth-year student Jean Rafael López carried out structural simulations of several channel designs using specialized software, which changed our philosophy about building the channel: it is essential to use a steel structure to support the acrylic panels of the channel. With other materials, the mechanical deformations would be unacceptable.

This meant obtaining substantial quantities of steel L-section and square-section profiles, which was done through a process of redistributing some consumable materials acquired through the project, and a donation of materials from IMRE (University of Havana).

The construction process began in the second half of September and is advancing at a good pace, partly thanks to the blacksmithing skills of the fifth-year Physics student mentioned above.

One of the first steps to set the process in motion was the relocation and major repair of the small glass-walled wave channel, in which Marlon Mederos cooperated substantially, as did Luis Abel Rodríguez. This will ensure that the experiments are not interrupted while the large channel is being built.



Fig. 4. A new discipline emerges in Physics: channelculture. Left panel: the old “small” wave channel is moved and repaired to make room for the future “large” channel. Center and right panels: the fifth-year Physics undergraduate cutting (center) and welding (right) steel profiles that will serve as the structure for building the new wave channel planned in the project.

Task 5 Once again, the Summer School for pre-university students gets close to the ARES project

As every summer, the Faculty of Physics held a Summer School with vocational goals. There, Luis Abel Rodríguez led a team of pre-university students around the project “The secret of the tilt: how an angle decides where particles float in the sea”, and Daniela Martínez led the project “Optical calibration of the PlanktoScope: measuring magnifications, thicknesses and biodiversity”. This group won Second Place in the Tenth-Grade Students category at the School.

Task 6 The FabLab takes off: “Kick off FabLabComplex” Micro-workshop in August 2026

Although the equipment and materials needed to set up a digital fabrication workshop at the Faculty of Physics of the University of Havana arrived successfully — as documented above — we went one step further. Amid the country's precarious economic situation — in particular, the permanent electricity deficit — we showed that the digital fabrication laboratory funded by ARES was partially functional less than a month after the equipment arrived.

This bet was based on the fact that the Physics building shares a circuit with the “Calixto García” hospital and suffers much less from the lack of electrical power. Specifically, we called a “Micro-workshop” entitled “Kick off FabLabComplex”, held on August 10–12, 2026, in the middle of the vacation period — the cooperation of the Faculty of Physics leadership in keeping the Physics building open was a fundamental element of the task. In the workshop, three groups were organized to build, from scratch, three pieces of equipment relevant to the objectives of the ARES project. The first, led by Orestes Chávez, would build an anemometer. The second, led by Alex Manuel Rivera, a turbidity meter. The third, led by Luis Abel Rodríguez, a water-depth gauge for a channel. These groups included Physics students as participants, as well as engineers, young biologists, a physician and a social scientist, which made for a heterogeneous and inclusive group in every sense.

Despite some organizational hiccups, everyone's work and, in particular, the extraordinary final push by Alex Manuel Rivera meant that on the last day of the workshop, at 8 PM, all three projects were successfully completed.





Fig. 5 “KickoffComplex” Micro-workshop (August 10–12, 2026). In the four left-hand panels, various moments of the work of the different teams can be seen. In the right-hand panel (taken at 8 PM on Wednesday, August 12, 2026), the final group of “survivors” who saw the three workshop projects completed, which are shown below in the same photo: turbidity meter (left), anemometer (center) and water-level gauge (right).

Task 6 Academic actions

PhD candidate Luis Abel Rodríguez de Torner gave a doctoral progress seminar at the Center for Complex Systems of the UH on September 28, 2026. Earlier, Luis Abel took part for a week in the 5th Training School on Wave-Structure Interaction (Florence, September 14–18, 2026), funded by the ARES project.

In addition, a number of articles associated with the “exact and natural” sciences are being worked on, summarized in the figure below:

Microorganisms as indicators of marine pollution: challenges for <i>in situ</i> monitoring Daniela Martínez-Ortiz^{1,2}, Alex M. Rivera-Rivera^{2,3}, Aramis Rivera⁴, Denis Terwagne^{5,6}, Alexandre Campo⁷, Luis A. Rodríguez-de-Torner^{1,2}, Ernesto Altshuler^{1*}, Stamatios C. Nicolis^{7*}
Low-Cost Spectrophotometers: A Comparative Evaluation of Open-Source Architectures Alex M. Rivera-Rivera^{1,2}, Daniela Martínez-Ortiz^{3,2}, François Quitin⁴, Denis Terwagne^{5,6}, Stamatios C. Nicolis⁷, Ernesto Altshuler^{3*}, Alexandre Campo^{7*}
An Accessible Fluorometric Protocol for Chlorophyll <i>a</i> Quantification Using Open-Source Spectral Sensors D. Martínez-Ortiz^{3,2}, A. M. Rivera-Rivera^{1,2}, A. Campo⁷, D. Terwagne^{5,6}, F. Quitin⁴, E. Altshuler³, S. C. Nicolis^{1,*}
Low-Cost Open-Source Absorbance Spectrophotometry on a Monochrome Global-Shutter Camera Alex M. Rivera-Rivera^{1,2}, Daniela Martínez-Ortiz^{3,2}, Dayaris Hernández⁸, Aramis Rivera⁸, François Quitin⁴, Denis Terwagne^{5,6}, Stamatios C. Nicolis⁷, Ernesto Altshuler³, and Alexandre Campo⁷
Frugal Science for Marine Monitoring: The Modular PlanktoScope—A Reproducible, Low-Cost Microscope for <i>In Situ</i> Detection of Biological Indicators D. Martínez-Ortiz^{3,2}, A. M. Rivera-Rivera^{1,2}, A. Cornu⁸, S. C. Nicolis⁷.

- *Microorganisms as indicators of marine pollution: challenges for in situ monitoring* — Submitted to “Environmental Modelling and Assessment” in April 2026. Still awaiting the response of an anonymous referee in the first round of review.
- *Low-Cost Spectrophotometers: A Comparative Evaluation of Open-Source Architectures* — In final review by the authors, to be submitted to “Review of Scientific Instruments”.
- The other three articles (chlorophyll *a* fluorometric protocol; absorbance spectrophotometry on a monochrome global-shutter camera; modular PlanktoScope) — In preparation.

Fig. 6 Articles in progress, within the topics of exact sciences, natural sciences and engineering.

The academic exercises carried out by the social-sciences PhD candidate are included in the next section.

Task 6. Continuing the research within the social sciences

1. Continuation of the doctoral stay of Mayelín García-Remus at the Université de Louvain-la-Neuve (February–August 2026).
2. Participation of the PhD candidate in the training spaces at the Interdisciplinary Research Center on Work Organization and Society (CIRTES), where she is doing her internship, where she took part especially in the following activities:
 - a) Participation in the 10th edition of the EMES International PhD School on Social Economy, which aimed to investigate and discuss epistemologies and methodologies related to and identified with the social economy, where the following lectures were given: (i) “How can research on solidarity-based social enterprise contribute to fostering economic justice in the European context?” by Luciane Lucas dos Santos, full professor and researcher at the Centre for Social Studies of the University of Coimbra, Portugal, and Fernanda Wanderley, Director of the Institute for Socioeconomic Research, Bolivian Catholic University “San Pablo”, Bolivia. (ii) “Power in just transitions for sustainability: Who, what, how and why?” by Francesca Petrella, Professor at Aix-Marseille University.
3. The drafting of the article *Community and citizen participation in water-quality monitoring. A systematic review* was completed; it aims to review the scientific literature to analyze the forms of citizen and community participation in water-quality monitoring, and the use of low-cost tools for monitoring.
4. Meeting with the two supervisors to analyze the progress of the project and the most relevant findings of the literature review carried out. It resulted in the proposed structure of the thesis, based on 3 articles:
 - Article (theoretical foundation): What does the literature tell us about the conditions, motivations and forms of citizen and community participation in monitoring water pollution, and what lessons does it offer for the validation and implementation of low-cost monitoring tools, with different community actors?
 - Article 2: The Casablanca community and the monitoring of water pollution in Havana Bay. This is the empirical core. How do the different actors of the communities under study interact with the new technology, or how do they come to know and question their exposure?
 - Article 3. Reflection on the interdisciplinary dialogue around the new technology and the knowledge of the different social actors who interact with it. How does a shared object make a problem (water pollution) visible and public, and how do actors from different disciplines negotiate around it?
5. Meeting with Florence Degavre to present the first version of the article and the activities to be carried out during the next internship.
6. Meeting with the president of the Casablanca people's council to continue the application of the surveys.



Fig. 6 Members of the “Casablanca” people’s council community filling out our surveys.

7. Working session with the director of the CIMAB laboratory, to coordinate the tasks to be carried out for the validation of the equipment that will be used in the communities.
8. Participation in the FabLab organized in August by the Faculty of Physics, from which interviews with key informants about digital fabrication in Cuba were conducted and a proposed instrument to measure social appropriation of technologies was designed. Its application, for validation, is pending.
9. Preparation and submission, to the scientific committee of the International Seminar “Intersectionality, Equity and Social Policies”, of the proposal for a panel on Technologies and equity gaps. It was approved and will be presented next November.
10. Methodological meeting on possible social scenarios and alternative ways of working with the Casablanca community from the citizen-science paradigm, based on the progress in the design of the tools being developed by the PhD candidates.

Conclusion

Despite the dramatic economic situation in the country, which implies a serious deterioration in the living conditions of the participants, null access to the bay’s buoys and zero maintenance of the infrastructure through institutional channels, the project is at two moments of clear take-off: (a) the organization of a digital fabrication laboratory (FabLabComplex) and (b) the start of the construction of the wave channel.

Dr. C. Ernesto Altshuler
Project Director, Cuban side